DATABASE MANAGEMENT | SQL PROJECT

E-LEARNING PURCHASE ANALYSIS

Database Design, SQL Implementation and Business Insights

PROJECT AREA	Relational Database Management System
DATABASE	elearning
SQL DIALECT	MySQL
SUBMITTED BY	_____

Submitted in partial fulfilment of the SQL end-module assignment

1. Project Overview

Purpose, objectives and scope of the database exercise

This project develops a small relational database for an e-learning platform. The database records learners, the courses offered by the platform, and the purchases made by learners. SQL statements are used to create the database structure, insert sample records, connect related tables and produce summaries that describe purchasing activity.

The work moves from database setup to analysis. It demonstrates how related information can be stored separately and then combined when a report is required. The sample data represents five learners, five courses and eight purchase transactions dated in July 2026.

Project objectives

- Create and select a database named elearning.
- Design three related tables with primary-key and foreign-key constraints.
- Insert sample learner, course and purchase records.
- Use INNER JOIN, LEFT JOIN and RIGHT JOIN to retrieve related information.
- Apply calculated columns, formatting, sorting, grouping, aggregate functions and filtering.
- Identify purchasing patterns, course demand and courses without purchases.

Scope and approach

The report is based on the supplied SQL script. It covers the schema and the queries included in that file; it does not assume additional platform features such as logins, payments, course progress or refunds. The examples use MySQL syntax, including the FORMAT() function.

Project workflow

Stage	Work performed
1. Setup	Create the elearning database and select it for use.
2. Structure	Create learners, courses and purchases tables.
3. Data entry	Insert the sample records into each table.
4. Retrieval	Join tables to display learner and purchase details.
5. Analysis	Summarise spending, sales quantities and category activity.

The project is a focused SQL practice exercise: its value lies in showing how a relational structure supports accurate retrieval and basic business reporting.

2. Database Design and Sample Data

Relational structure and the records used for analysis

The database contains three tables. Each table has a primary key that uniquely identifies its records. The purchases table connects learners and courses through foreign keys, allowing a purchase to be associated with one learner and one course.

Table	Primary key	Important columns	Role
learners	learner_id	full_name, country	Stores learner details.
courses	course_id	course_name, category, unit_price	Stores course information and price.
purchases	purchase_id	learner_id, course_id, quantity, purchase_date	Records each purchase.

Relationships

- learners.learner_id is referenced by purchases.learner_id.
- courses.course_id is referenced by purchases.course_id.
- A learner can have multiple purchase records; each purchase refers to one learner.
- A course can appear in multiple purchase records; each purchase refers to one course.

Sample data volume

Table	Number of records	Example values
learners	5	Arun Kumar; Priya Sharma; John Smith; Maria Garcia; David Lee
courses	5	Python Programming; Data Analytics; SQL for Beginners; Power BI Basics; Machine Learning
purchases	8	Purchase IDs 1–8, dated 1–18 July 2026

Table creation pattern

```
CREATE TABLE purchases (
  purchase_id INT PRIMARY KEY,
  learner_id INT,
  course_id INT,
  quantity INT,
  purchase_date DATE,
  FOREIGN KEY (learner_id) REFERENCES learners(learner_id),
  FOREIGN KEY (course_id) REFERENCES courses(course_id)
);
```

The foreign-key definitions help maintain referential integrity: a purchase cannot refer to a learner or course that does not exist in the referenced table. The supplied script also uses DESCRIBE statements to inspect the structure after table creation.

Course prices are stored as DECIMAL(10,2), which is suitable for values that require two decimal places. Purchase quantity is stored as an integer, and purchase_date uses the DATE data type.

3. SQL Retrieval and JOIN Operations

Combining learner, course and purchase information

The central retrieval query joins all three tables. The learners table supplies the learner name, purchases supplies quantity and date, and courses supplies the course name, category and unit price. The calculated amount is quantity multiplied by unit_price.

```
SELECT l.full_name, c.course_name, c.category,
    p.quantity,
    p.quantity * c.unit_price AS total_amount,
    p.purchase_date
FROM learners AS l
INNER JOIN purchases AS p
 ON l.learner_id = p.learner_id
INNER JOIN courses AS c
 ON p.course_id = c.course_id;
```

What the JOIN queries demonstrate

Operation	Purpose in this project	Observed result
INNER JOIN	Returns matching learner, purchase and course records.	8 purchase rows.
LEFT JOIN	Keeps every learner, even if no purchase exists.	All 5 learners appear; each has purchases in this data.
RIGHT JOIN	Keeps every learner from the right-hand table.	Same learner coverage in the supplied query.

Purchase detail summary

Learner	Course	Qty	Amount
Arun Kumar	Python Programming	1	499.00
Arun Kumar	Data Analytics	1	699.00
Priya Sharma	SQL for Beginners	2	798.00
John Smith	Data Analytics	1	699.00
John Smith	Machine Learning	1	899.00
Maria Garcia	Power BI Basics	2	1,198.00
David Lee	Python Programming	1	499.00
David Lee	Data Analytics	2	1,398.00

The script also uses FORMAT(amount, 2) to display calculated amounts with two decimal places. An ORDER BY clause sorts the purchase details from the highest amount to the lowest. In this dataset, David Lee's Data Analytics purchase is the largest single transaction at 1,398.00.

The LEFT JOIN and RIGHT JOIN queries are useful demonstrations of outer joins. Because every sample learner has at least one purchase, neither query produces a learner row with NULL purchase fields.

4. Aggregations and Business Analysis

Using GROUP BY, aggregate functions and sorting

The analysis queries move beyond individual transactions and calculate totals at learner, course and category level. SUM() adds monetary values or quantities, COUNT(DISTINCT ...) counts unique learners or categories, and GROUP BY defines the level at which each summary is produced.

Total spending by learner

Learner	Country	Total spending
David Lee	Singapore	1,897.00
John Smith	USA	1,598.00
Arun Kumar	India	1,198.00
Maria Garcia	Spain	1,198.00
Priya Sharma	India	798.00

The total value of all eight transactions is 6,689.00. David Lee has the highest cumulative spending in the sample, followed by John Smith.

Course sales quantity

Course	Units sold
Data Analytics	4
Python Programming	2
SQL for Beginners	2
Power BI Basics	2
Machine Learning	1

Data Analytics has the highest quantity sold, with four units. The SQL statement returns the top three rows by quantity, but three courses are tied at two units for the remaining positions. Since the query does not add a secondary sort key, the order among tied courses is not guaranteed.

Revenue by category

Category	Revenue	Unique learners
Data Science	3,695.00	3
Data Visualization	1,198.00	1
Programming	998.00	2
Database	798.00	1

Data Science contributes the largest revenue in the sample. It combines Data Analytics revenue of 2,796.00 and Machine Learning revenue of 899.00.

Technical note: the supplied category query orders by the alias total_revenue after applying FORMAT(). In MySQL, FORMAT() returns formatted text. For consistently numeric sorting, ORDER BY SUM(p.quantity * c.unit_price) DESC is preferable.

5. Findings, Recommendations and Conclusion

Interpretation of results and practical improvements

Key findings

- The database successfully represents five learners, five courses and eight purchases across three related tables.
- The total purchase value in the sample is 6,689.00.
- David Lee records the highest total spending at 1,897.00.
- Data Analytics is the most purchased course by quantity, with four units sold.
- Data Science is the highest-revenue category, generating 3,695.00.
- Arun Kumar and David Lee each purchased courses from more than one category.
- Every course appears in the purchase table, so the final LEFT JOIN query returns no unused courses.

Category diversity query

```
GROUP BY l.learner_id, l.full_name
HAVING COUNT(DISTINCT c.category) > 1;
```

The HAVING clause filters grouped results after the category count has been calculated. With the supplied records, the query identifies Arun Kumar and David Lee, each of whom purchased from two distinct categories.

Recommendations

- Add a secondary ORDER BY column, such as course_name, when ranking courses with tied sales quantities.
- Keep numeric values numeric for sorting and apply FORMAT() only for presentation in the final output.
- Consider adding NOT NULL constraints to essential fields such as names, course names, prices and purchase quantities.
- For a larger system, consider adding checks for positive prices and quantities, plus indexes on foreign-key columns.
- Extend the dataset with more transactions and dates before drawing wider conclusions about learner behaviour or course performance.

Conclusion

This project demonstrates the core stages of a relational SQL workflow: creating a database, defining linked tables, inserting records, retrieving joined information and producing grouped summaries. The results show how transaction-level data can be converted into useful measures such as learner spending, course sales and category revenue. The exercise also illustrates the importance of choosing appropriate joins, grouping columns and sorting expressions. Overall, the supplied SQL script provides a practical foundation for building more detailed e-learning reports.

Source basis

Prepared from the supplied file, SQL_End Module Assignment (1).sql. Numerical summaries in this report are calculated from the INSERT statements in that script. No external data has been added.